Supplementary material

Brand-constrained diffusion for broadcast packaging with a late-window temporal residual

This supplement holds the configuration tables and the full numeric results that were condensed in the main paper. Nothing here changes a claim in the main text. Table numbering follows the supplementary series S1 to S5.

S1. Temporal residual configuration

Table S1. Layer configuration of the temporal residual. Each gated call evaluates the module once on the full 16 frame latent sequence.

Layer	Configuration	Output shape	Parameters
Conv3d (in)	4 in, 16 out, kernel 3x3x3, padding 1	16x16x32x32	1,744
GroupNorm	4 groups, 16 channels	16x16x32x32	32
Time MLP	input 1, hidden 32, output 32, SiLU	2 x 16 (scale, shift)	1,120
Scale and shift	features times (1 + scale) plus shift	16x16x32x32	0
Conv3d (out)	16 in, 4 out, kernel 3x1x1, padding (1,0,0)	16x4x32x32	196
Total			3,092

S2. Conditions

Table S2. All conditions used in the study. The locked setting is a deterministic bound and not a competing generator.

Condition	Adapter	Residual	Role
No LoRA	off	none	Baseline without style adaptation
LoRA	on	none	Main baseline for the gate comparison
Locked	on	background held fixed	Deterministic bound only
Full injection	on	all 11 calls	Unrestricted residual use
Pre-registered gate	on	last 3 calls	Setting carried over from the submission
Wide gate	on	last 8 calls	Sweep optimum on this split
No adapter, wide gate	off	last 8 calls	Residual without style adaptation
LoRA, no Canny	on	none	Layout condition removed

S3. Full gating schedule

Table S3. Every tested value of K. The change column is baseline minus treatment, so a positive value means lower jitter with the residual. The p column is unadjusted.

Gated calls	Background jitter	Change	95% CI	p	Pairs agreeing	Relative
0	0.2676 ± 0.0100	—	—	—	—	—
1	0.2664 ± 0.0103	+0.00116	0.00012 to 0.00219	0.039	17/30	+0.44%
2	0.2669 ± 0.0103	+0.00070	−0.00033 to 0.00168	0.191	18/30	+0.26%
3	0.2653 ± 0.0105	+0.00226	0.00118 to 0.00331	3.7 x 1e-4	22/30	+0.84%
4	0.2651 ± 0.0112	+0.00245	0.00125 to 0.00367	6.3 x 1e-4	22/30	+0.91%
6	0.2614 ± 0.0108	+0.00618	0.00435 to 0.00795	1.0 x 1e-5	27/30	+2.31%
8	0.2597 ± 0.0114	+0.00790	0.00530 to 0.01062	1.0 x 1e-5	26/30	+2.95%
11	0.2786 ± 0.0088	−0.01106	−0.01361 to −0.00856	1.0 x 1e-5	2/30	−4.13%

S4. Alternative measures

Table S4. Alternative measures for the pre-registered gate against independent sampling. SSIM is higher when better, the other rows are lower when better, and differences are oriented so that a positive value favours the gate.

Measure	Independent	Gate, K = 3	Difference	95% CI	p
Background jitter	0.2676	0.2653	+0.00226	0.00118 to 0.00331	3.7e-4
Adjacent frame SSIM	0.2423	0.2548	+0.01257	0.00972 to 0.01550	1.0e-5
High pass jitter	0.05785	0.05721	+0.00064	0.00011 to 0.00117	0.029
Worst frame pairs	0.3425	0.3411	+0.00146	−0.00050 to 0.00324	0.146
Static region	0.2635	0.2612	+0.00228	0.00122 to 0.00330	3.2e-4
Moving region	0.3215	0.3186	+0.00281	0.000003 to 0.00563	0.063

S5. Reproduction notes

All numbers come from four runs. The formal run holds 150 sequences, ten held-out assets by three seeds by five conditions. The schedule sweep adds 180 sequences over six K values. The attribution run adds 60 sequences for two conditions. The static region analysis reuses the stored frames. Every condition shares the same three seeds, initial noise,

Canny condition and prompt, and the unit of analysis is the asset and seed pair, so n = 30. Bootstrap intervals use 5,000 resamples, sign-flip tests use 100,000 draws and the resampling seed is 20260903. The sweep reproduces the single sample value for K = 3 exactly, 0.2649148503939311, and the offline metric pipeline reproduces the stored jitter to within 2.5e-7. Removing the Canny condition makes the output independent of the asset, which we confirmed by comparing stored frames across assets, so that condition is reported descriptively.